Tonne-Mile Demand Model

How much of the move in the Baltic Dry Index is cargo volume, haul distance, fleet growth and port congestion? A decomposition built only from public data, Dec 2012 to Sep 2025.

Summary

Dry bulk freight rates are set by the balance between demand for tonne-miles and the effective supply of ships. This note measures both sides from public sources and attributes changes in the Baltic Dry Index to four forces: cargo volume on 271 major bulk routes, the average distance that cargo sails, the size of the bulk carrier fleet, and the share of fleet time absorbed in port. Over 154 months a one percent rise in tonne-miles relative to effective fleet has been associated with a 2.4 percent rise in the index (standard error 0.8, estimated on twelve-month changes); the relationship explains 7% of the variance of twelve-month changes in the log index. Over rolling twelve-month windows the four physical forces leave about 91% of a typical move unexplained (mean residual 0.39 log points against a typical move of 0.43). Two public market-price signals, the Brent crude price and the iron ore price three months earlier, carry much of the missing commodity cycle: with them the unexplained share falls to 55% in sample and 62% out of sample (against 94% for the physical forces alone). They are reported as separate market-signal bars, not as physical forces. In the latest window (Sep 2024 to Sep 2025) the index moved -4 percent; the largest physical contribution was fleet growth (-0.07 log points) and the residual was +0.20.

The question and the identity

Tonne-mile demand is the sum over routes of tonnes shipped times sailing distance; average haul is tonne-miles divided by tonnes, so the change in tonne-miles is exactly the change in volume plus the change in haul. Fleet supply is bulk carrier deadweight in service. Effective supply multiplies the fleet by the share of time at sea, derived from median port time per call and calls per ship. For any two months the identity log(BDI1/BDI0) = e[dlog V + dlog H] minus e[dlog F + dlog S] + residual holds by construction, where e is a single elasticity estimated by regressing twelve-month changes in the log index on twelve-month changes in the log of tonne-miles over effective fleet. Endpoints are three-month averages. The residual is reported as the fifth component; it is never spread across the others.

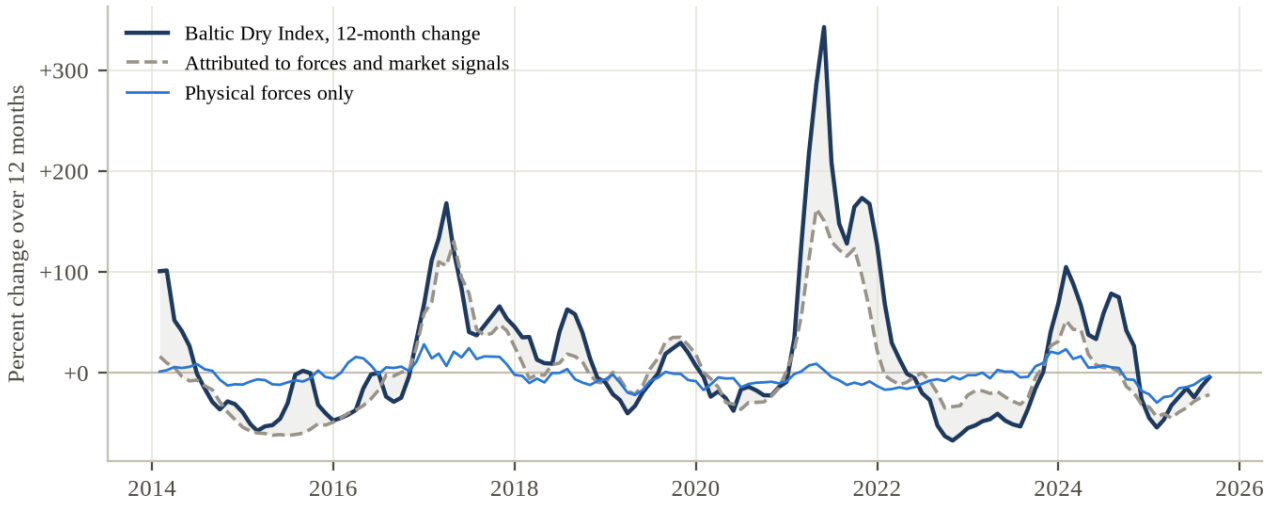


Figure 1. Twelve-month percent change in the Baltic Dry Index and the part attributed to cargo volume, haul, fleet and port time plus the two market-price signals at the estimated elasticity (2.36); endpoints are three-month averages. Shaded area is the residual. Sources: Statistics and Research Aland (index, originator Baltic Exchange); UN Comtrade, Brazil Comex Stat, UNCTADstat, IMF PortWatch (port calls, chokepoint transits) and this project's distance matrix from NGA World Port Index ports (inputs); World Bank Pink Sheet (prices).

Data

Series	Source	Frequency and coverage	Retrieved
Index of international dry bulk freight (Baltic Dry Index, 1985 = 1 000) 2011-2025 by month	Statistics and Research Aland (ASUB), official statistics of Aland	monthly; 2011-01 to 2025-12	2026-10-20

Series	Source	Frequency and coverage	Retrieved
Brazil Comex Stat: monthly exports by destination country for iron ore, soybeans, maize, soybean meal, sugar, manganese ore, bauxite and wood chips	Ministerio do Desenvolvimento, Industria, Comercio e Servicos (MDIC), Brazil	monthly; 1997-01 to 2026-09 (last update 2026-10-06)	2026-10-20
Index of global real economic activity (Kilian index), based on dry cargo ocean freight rates	Federal Reserve Bank of Dallas (index by Lutz Kilian)	monthly; 1968-01 to 2026-09	2026-10-20
IMF PortWatch: daily port activity and chokepoint transits (AIS-based)	International Monetary Fund and University of Oxford (PortWatch)	daily (aggregated here to months and to global daily totals); 2019-01-01 to 2026-10-09, 2065 ports, 28 chokepoints	2026-10-20
World Port Index (Pub. 150), port locations and characteristics	US National Geospatial-Intelligence Agency (NGA), Maritime Safety Information	updated continuously; snapshot at retrieval; about 3808 ports worldwide	2026-10-20
OECD Composite Leading Indicators, amplitude adjusted (long-term average = 100)	Organisation for Economic Co-operation and Development (OECD)	monthly; 2005-01 to 2026-09, 22 areas	2026-10-20
Consumer price index, United States, all items (OECD Prices database)	OECD (source: US Bureau of Labor Statistics)	monthly; 1960-01 to 2026-09, except 2025-10	2026-10-20
UN Comtrade: monthly bilateral trade in iron ore, coal and lignite, grains and soybeans, and minor bulks (net weight)	United Nations Statistics Division (UN Comtrade)	monthly from 2010; annual from 2000 (annual unit values); 2010-01 to latest published month per reporter; 4 reporter groups	2026-10-20
UN Comtrade: annual exports of iron ore, coal and grains to every partner since 1980 (net weight)	United Nations Statistics Division (UN Comtrade)	annual; SITC Revision 2 1980 to 1995; HS as reported 1988 to 2025	2026-10-20
UNCTADstat maritime transport statistics: merchant fleet, ships built and scrapped, port calls, seaborne trade	United Nations Conference on Trade and Development (UNCTAD), UNCTADstat Data Centre	annual; US.MerchantFleet: 1980 to 2026; US.ShipBuilding: 2014 to 2025; US.ShipScrapping: 2014 to 2025; US.PortCalls: 2018 to 2023; US.SeaborneTrade: 2000 to 2024; US.PortCalls_S: 2018S01 to 2023S02; US.PortCallsArrivals: 2018 to 2023	2026-10-20
World Bank Commodity Markets Outlook, monthly prices (Pink Sheet)	World Bank	monthly; 1960-01 to 2026-09, nominal US dollars	2026-10-20

Routes are measured from the customs data of the exporter wherever it publishes timely monthly data (Australia, Brazil, Indonesia, South Africa, United States, Canada, Ukraine, Argentina, Colombia) and from the importer for Russian and Mozambican origins (India, Japan). Chinese import records lag by more than a year and are used only for cross-checks. Distances are volume-weighted blends between representative terminals on a maritime routing network, with iron ore and most coal routed around the Panama Canal (Capesize tonnage; coal from the Americas, largely on Panamax ships, takes the canal), Suez routes lengthened towards the Cape in proportion to the fall in dry bulk transits through Bab el-Mandeb observed by IMF PortWatch since November 2023, and Panama routes lengthened the same way during the 2023 to September 2024 draught restrictions. The three largest routes carry 48% of covered tonne-miles. Validation: 31 checks passed, 6 warnings, 0 failures (full report on the site).

Results

Between the first and last twelve months of the panel, tonnes on the covered routes grew +28% and tonne-miles +30%, implying an average haul change of +1.4%; the bulk carrier fleet grew +48% to 1,057 million dwt. Figure 2 attributes every rolling twelve-month change of the log index. Fleet growth is a steady negative contribution; volume and haul alternate as the main positive drivers; the port-time factor matters in 2021 and 2022; the residual carries the spikes.

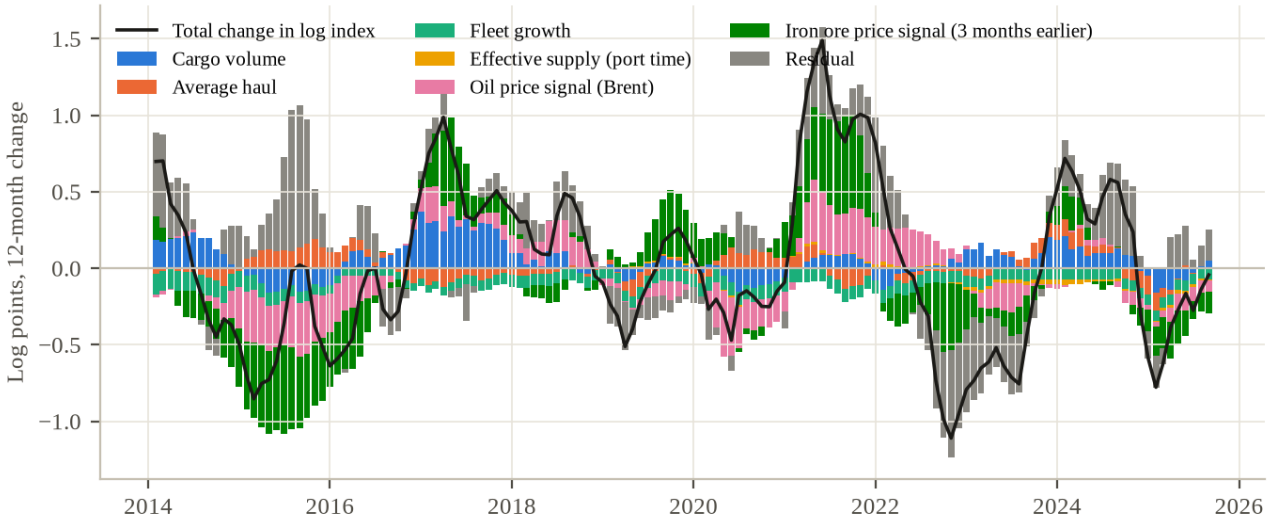


Figure 2. Twelve-month change in the log Baltic Dry Index (line) and its attribution (bars), log points. Endpoints are three-month averages.

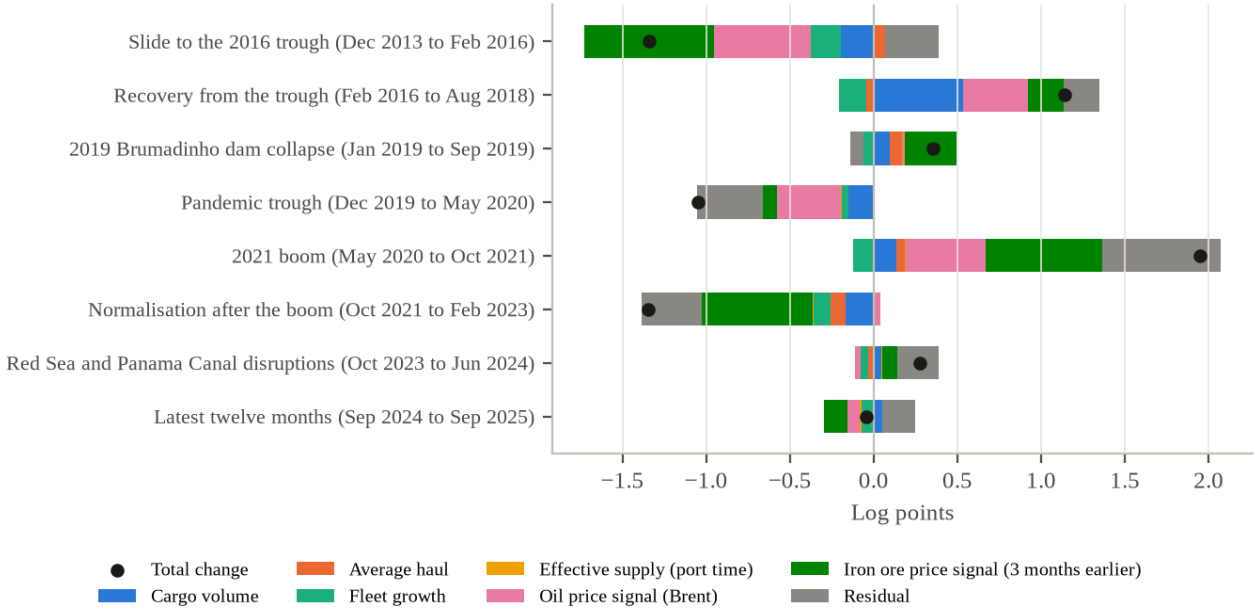


Figure 3. Named episodes covered by the panel: contributions in log points (bars) and the total change (dot). An episode is listed only when both endpoints fall inside the data.

Episode	Window	Total	Vol.	Haul	Fleet	Port	Oil	Ore	Resid.	Explained
Slide to the 2016 trough	Dec 2013 to Feb 2016	-1.34	-0.20	+0.07	-0.18	-0.00	-0.58	-0.77	+0.32	124% (23%)
Recovery from the trough	Feb 2016 to Aug 2018	+1.14	+0.54	-0.04	-0.16	-0.00	+0.39	+0.21	+0.21	81% (29%)
2019 Brumadinho dam collapse	Jan 2019 to Sep 2019	+0.35	+0.09	+0.07	-0.06	+0.01	+0.01	+0.31	-0.08	123% (32%)
Pandemic trough	Dec 2019 to May 2020	-1.05	-0.15	+0.00	-0.04	-0.01	-0.38	-0.09	-0.39	63% (18%)

Episode	Window	Total	Vol.	Haul	Fleet	Port	Oil	Ore	Resid.	Explained
2021 boom	May 2020 to Oct 2021	+1.95	+0.14	+0.05	-0.12	+0.01	+0.48	+0.69	+0.71	64% (4%)
Normalisation after the boom	Oct 2021 to Feb 2023	-1.34	-0.17	-0.09	-0.10	-0.01	+0.04	-0.67	-0.36	74% (27%)
Red Sea and Panama Canal disruptions	Oct 2023 to Jun 2024	+0.28	+0.05	-0.03	-0.05	+0.00	-0.03	+0.09	+0.25	10% (-10%)
Latest twelve months	Sep 2024 to Sep 2025	-0.04	+0.05	-0.00	-0.07	-0.01	-0.07	-0.14	+0.20	n/m

Table 1. Episode decomposition in log points. Vol. is cargo volume and Port is port time; Oil and Ore are the market-signal bars (Brent, and iron ore three months earlier). Explained is one minus residual over total, with the physical-forces-only share in brackets; it can exceed 100 percent when the residual opposes the move and is shown as n/m for moves under 0.10 log points.

Market signals

The physical balance moves a few percent a year while the index moves by tens of percent. Two public World Bank prices carry much of the missing variation and are added after the four physical forces as separately named bars, with the elasticity of the identity unchanged: the Brent crude price over the same window (coefficient 0.49, t 2.8) and the iron ore price, CFR China, over the window ending three months earlier (0.83, t 4.2); the lead keeps the freight contained in a delivered price out of the same window. The coefficients come from a regression without intercept of the physical residual on the two price changes over every rolling window. In sample the unexplained share of a typical move falls from 91% to 55%. Out of sample, refitting every coefficient each month with data that ended at least twelve months earlier (69 months from Jan 2020 to Sep 2025), R squared against a zero forecast rises from 0.06 to 0.62 and the unexplained share falls from 94% to 62%; a benchmark with no fundamentals, the index's own twelve-month change three months earlier, reaches 0.46. These are market signals, not physical forces: oil and iron ore prices react to the same global and Chinese demand shocks as freight, and other industrial commodity prices carry similar information, so the bars measure a commodity cycle that the customs-based balance misses rather than a causal channel. The two series were chosen after testing several candidate prices and leads on the same sample, which flatters the out-of-sample gain.

Further additions were scored with the same protocol (headline out-of-sample R squared 0.62). Adding minor bulks (sugar, soybean meal, cement, manganese and bauxite ores, petroleum coke, potash, wood, scrap) for the 9 exporters already covered (about 195 million tonnes a year, a subset of world minor-bulk trade), about 9% more tonne-miles, gives 0.62 (unexplained share 62% out of sample against 62%). The OECD leading indicators, China and the G20 as a third signal and China in place of iron ore, at leads of zero, three and six months (9 variants), change the out-of-sample R squared by -0.39 to +0.03; the best variant (China leading indicator added, 6 months earlier) gains +0.03. None clears the margin of 0.05 set to cover the choice of lead, so the headline keeps the major bulks and the two prices.

Four decades: 1980 to 2025

An annual version of the decomposition runs from 1980 to 2025 (45 annual changes). Tonnes come from the annual UN Comtrade records of the same exporters for every partner (SITC Revision 2 before HS), chain-linked on the series (exporter and commodity group) at least 90% measured in both years of each change (on average 84% of the year's sea-route tonnes); unit-value estimates for missing, estimated or understated weights fill the levels; the fleet is the UNCTAD series; the freight index is the Baltic Dry Index from 2011 and, before, a proxy: the Kilian index of dry cargo freight rates (Federal Reserve Bank of Dallas; Kilian 2009, Am. Econ. Rev. 99(3); Kilian 2019, Econ. Letters 178), converted back to nominal terms with US consumer prices and calibrated on 179 months where both exist (annual changes correlate at 0.994). Over the period tonne-miles on the covered trades rose 5.8-fold and the fleet 5.7-fold. On annual changes the elasticity is 3.27 (standard error 0.71); the forces leave 87% of a typical annual move unexplained, 79% with the two price signals (84% out of sample, 1996 to 2025). In the 2000s boom (2002 to 2007, +1.81 log points) volume contributed +1.47, haul -0.24, fleet -0.82, the price signals +0.90 and the residual +0.51. In the 2008 crash (2007 to 2009, -1.00 log points) volume contributed +0.43, haul -0.07, fleet -0.47, the price signals -0.17 and the residual -0.73. In the delivery wave (2010 to 2016, -1.45 log points) volume contributed +1.16, haul -0.11, fleet -1.47, the price signals -0.63 and the residual -0.40. Figure 4 shows every year (changes that use the proxy are shaded).

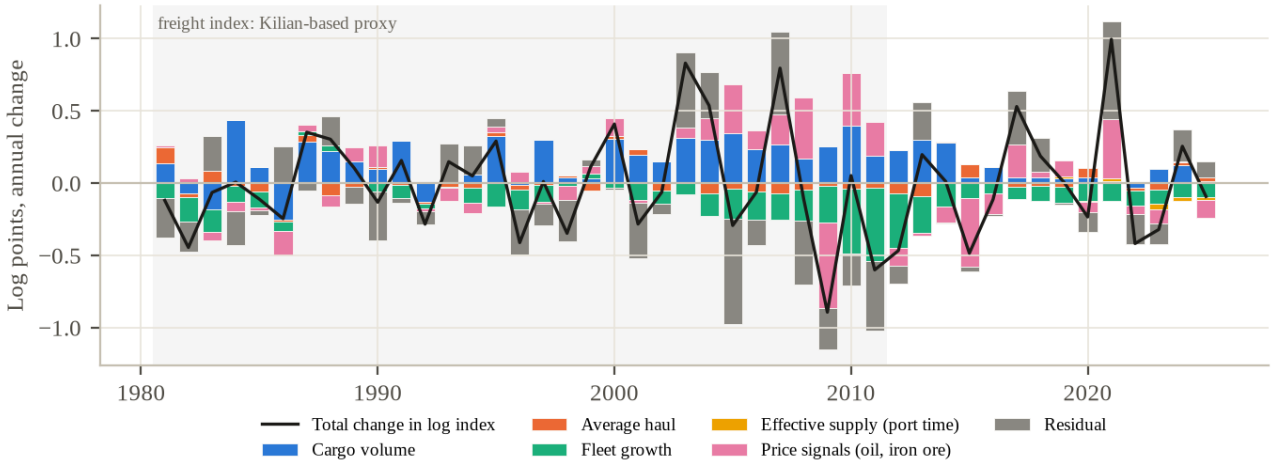


Figure 4. Annual change in the log of the annual average index (line) and its attribution (bars), log points; the price signals combine Brent (calendar-year average) and iron ore (twelve months to September). Shaded bars are changes that use the Kilian-based proxy for at least one of the two years. Sources: UN Comtrade (annual), UNCTADstat, IMF PortWatch, NGA World Port Index, Federal Reserve Bank of Dallas, OECD, ASUB, World Bank.

Robustness

The elasticity is re-estimated on alternative samples and transformations: twelve-month changes (headline, 142 observations): 2.36 (se 0.79, R squared 0.07, Durbin-Watson 0.53, intercept +0.05 log points a year, which the decomposition leaves in the residual); monthly levels with month dummies: -1.74 (se 0.81, R squared 0.20); monthly levels: 0.00 (se 0.53, R squared 0.00); quarterly averages, four-quarter changes: 4.56 (se 1.15, R squared 0.14); first half of the sample: 1.92 (se 0.90, R squared 0.07); second half of the sample: 3.04 (se 1.20, R squared 0.08). An unrestricted regression of monthly changes in the log index on the four input changes and their lags (152 months, R squared 0.20) gives contemporaneous coefficients of +1.88 (volume), -0.94 (haul), +16.68 (fleet) and +0.97 (port time), with standard errors 0.36, 0.93, 15.48 and 9.53; a Wald test of one common elasticity with the signs of the identity has p = 0.02. The levels regressions are reported for completeness only: both series trend and are strongly autocorrelated, so their coefficients are not reliable (low Durbin-Watson statistics); freight rates also move with information the inputs do not carry.

Limitations

Sailing speed, waiting outside ports and ballast ratios are not observable from public data and remain in the residual with bunker prices, sentiment and contract timing. Some customs records carry values without weights (United States monthly records before 2016); tonnes for those months are estimated from the reporter's annual unit values and flagged, and the Russia to Egypt grain route is left out because Egyptian records lack weights in most years. The route set covers the major bulks between the main exporters and importers; minor bulks are tested as an extension only, and regional trades that drive Supramax and Handysize rates are excluded. Representative ports approximate real terminals; the error is a few percent of a voyage and constant over time, so it affects levels rather than changes. Customs months record declarations, not sailings. Episodes shorter than twelve months compare endpoints in different seasons, and cargo volumes are seasonal, so their volume effect includes seasonal movement. The index composition changed in 2018. The freight index is republished by the Aland statistics office less often than the demand data, so the balanced panel ends earlier than the volume series. Deliveries and demolitions by ship type are not in the public UNCTAD tables; fleet growth is shown net.

Reproducibility

The pipeline (Python 3.11 or later) runs in six stages: ingest, validate, transform, model, export and report; one command rebuilds everything, unit tests check the distance routing, the tonne-mile aggregation and the adding-up of the decomposition, and a scheduled GitHub Actions workflow refreshes the data and republishes the site. Every raw file has a metadata record (source, retrieval time, coverage, licence, limitations, file hashes). Code, data and note: github.com/nathancouturier/Tonne-Mile-Demand-Model.